Ex:5
Date: 24.2.26

VERIFICATION OF NORTON'S THEOREM

AIM:

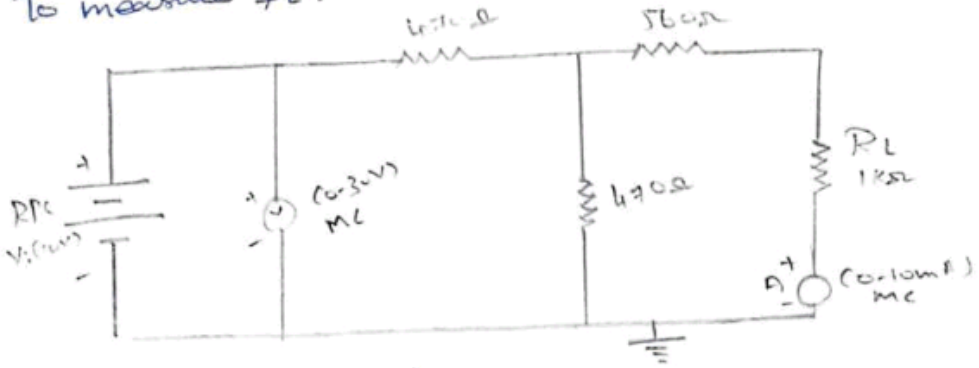
To Verify Norton's theorem, Practically and theoretically for the given DC Circuit.

APPARATUS REQUIRED.

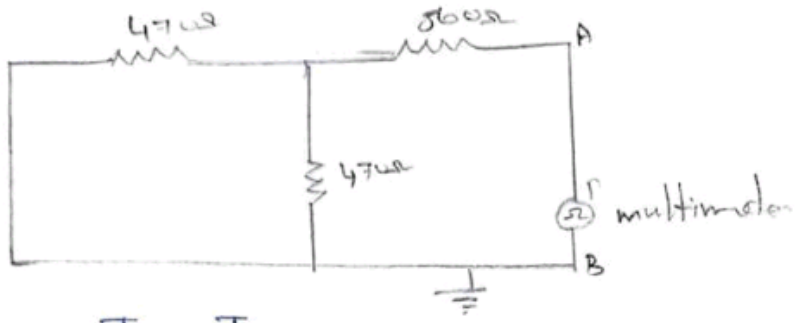
APPARATUS	SPECIFICATION	QUANTITY.
Regulated power Supply	(0-30V)	1
Voltmeter	(0-30V) MC	1
Ammeter	(0-10mA) MC	1
Resistors	470Ω, 560Ω, 1KΩ	2, 1, 1
Bread board	-	1
multimeter	-	1

CIRCUIT DIAGRAM:

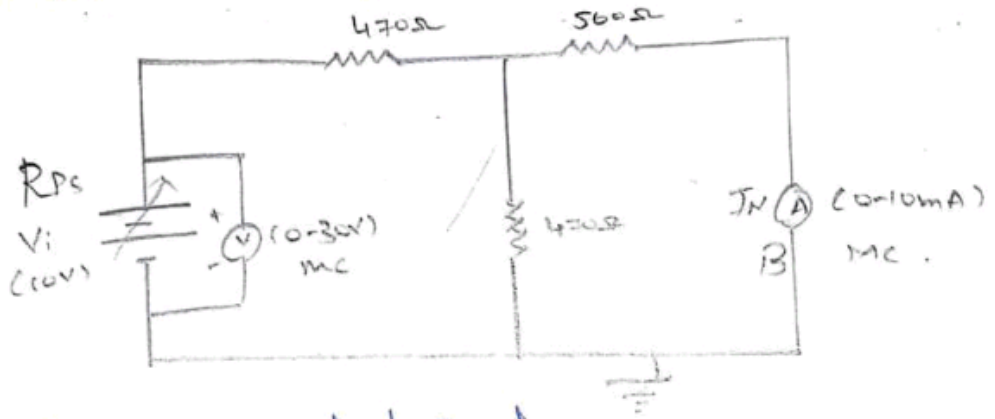
To measure I_L.



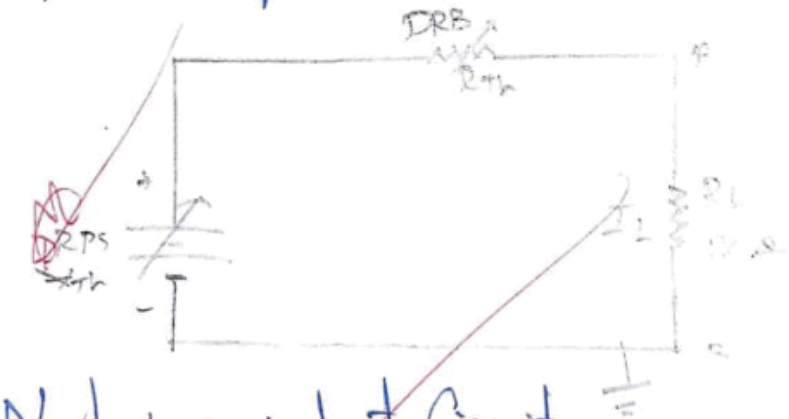
To measure R_{Th} or R_N



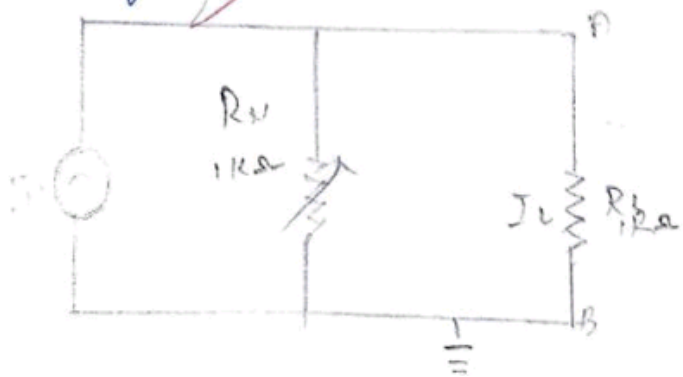
To measure I_N or I_{sc}



Thevenin's equivalent Circuit



Norton's equivalent Circuit.



Tabular Column: 2

To measure I_L

V_1 (Volts)	I_L (amps)
10V	2.67mA

Tabular Column: 2

To measure R_{Th} or R_N

V_1 (Volts)	R_{in} (Ω)
10V	7.6Ω

Tabular Column: 3

To measure I_N or I_{sc}

V_1 (Volts)	I_N (amps)
10V	5.94mA

model Calculation:-

Verification of Norton's theorem => $I_L = I_N * R_N/(R_N+R_L)$

Theoretical Calculation of I_L, I_N and Given Circuit.

$$R_T = 470 + \frac{470 \times 560}{470+560} \Rightarrow 470 + 255.5$$

$$R_T = 725.53\ \Omega$$

$$I_T = \frac{10}{725.53} = 0.013A$$

$$I_N = 0.013 \times \frac{470}{470+560} \Rightarrow 5.9mA$$

$$R_N = 560 + \frac{470 \times 470}{470+470} = 795\Omega$$

$$I_L = \frac{I_N \times R_N}{(R_N+R_L)}$$

$$= 5.9 \times 10^{-3} \times \frac{795}{795 \times 1000} = 2.01mA$$

0/8 verified

$$V_1 = 2.61 \times 10^{-3} \times 10^3$$

$$V_L = 2.61V$$

Verification.

$$I_L = \frac{I_N * R_N}{R_N + R_L} = 2.43mA$$

PROCEDURE:

1. Connect the Circuit as per the given Circuit diagram and set the RPS Input Voltage to 10V.

2. Measure and record the Voltmeter reading (V_1) and ammeter reading (I_1).

3. Switch off the Supply and measure the Thevenin/Norton equivalent resistance ($R_{TH} = R_N$).

4. Reconnect the Circuit to find the Thevenin Voltage ($V_{TH} = V_{OC}$) and note the Voltmeter reading.

5. Short the load to measure the Short Circuit Current ($I_{SC} = I_N$) and Verify Norton's theorem.

6. Calculate the I_L Native using the formula.

$$I_L = V_{th} / (R_{TH} + R_L)$$

Norton's theorem

$$I_L = I_n * R_N / (R_N + R_L)$$

7. theoretically Verify the Norton's theorem.

Description	Marks allotted	Marks obtained.
Experimental Setup / Connection	20	20
Observation / Reading	20	20
Calculation / Results	20	20
Viva	20	10
Record	20	20
	Total	90

90

RESULT

Thus Newton's theorem is verified practically and theoretically.